CO1-AT3 -TECHNICAL PRESENTATION
ITA-0609-MACHINE LEARNING

Ques5-Decision Tree Learning and Representation &
Ques1-Learning Problems and Perspectives

K.VISHWANATHAN

192312063

DR. SHRI VINDHYA A

QUESTION-5
Decision Tree and Its Representation

- **Key Concepts:**
Decision Tree Structure
Entropy
Information Gain
Pruning

- A **supervised Machine Learning algorithm** Used for **classification and decision-making**
Represents knowledge in a **tree structure**

- **Components:**
Root → Branch → Leaf

Entropy and Information Gain

- **Entropy**

- Entropy measures the **uncertainty or impurity** in a dataset.
- It helps determine how mixed the data is.

- **High Entropy** → Data contains more uncertainty.

- **Low Entropy** → Data is more organised or pure.

- **Information Gain**
- Information Gain measures how much uncertainty is reduced after splitting the data.
- It helps the algorithm select the **best feature** for making a decision.
- The feature with the **highest Information Gain** is selected as the decision node.

Working and Tree Pruning

Working of Decision Tree

- The algorithm starts with the complete dataset.
- It selects the best feature for splitting the data.
- The data is divided into different groups based on conditions.
- The process is repeated for each group.
- The process ends when a final prediction is obtained at a leaf node.
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Tree Pruning

- A very large tree may learn the training data too closely.
- This causes **overfitting**.
- Pruning removes unnecessary branches.
- It creates a simpler tree that performs better on new data.

Advantages, Limitations and Conclusion

Advantages

- Easy to understand and interpret.
- Simple to visualise.
- Useful for classification and decision-making.
- Can handle different types of data.
- Requires less data preprocessing.

Limitations

- Large trees can cause overfitting.
- Deep trees can become complex.
- Small changes in data may change the tree structure.
- A single decision tree may not always provide the highest accuracy.

Conclusion

- Decision Tree Learning selects the best features, divides data based on conditions, and reaches a final decision. **Entropy and Information Gain** help in selecting the best splits, while **pruning** helps reduce overfitting.

Question 2:
Concept Learning and Hypothesis Representation.

Concept Learning

- Concept Learning is the process of learning a general concept from training examples. The system identifies patterns from examples. Examples are usually classified as: **Positive examples** – belong to the concept

- **Negative examples** – do not belong to the concept

- The goal is to learn a rule that can classify new, unseen examples.

Concept Learning Task

- The learning system receives a set of training examples.
- Each example contains:
  - **Attributes/features**
  - **Target concept or class**
- The system analyses the examples and searches for a suitable rule.
- The learned rule is used to classify future examples.
- **Example:**
  Learning the concept **“Will Play Outside?”** using weather, temperature, and humidity.

Hypothesis Representation

- A **hypothesis** is a possible rule that explains the training examples.
- It represents the learner's current understanding of the concept.
- A hypothesis can be:
  - General
  - Specific
  - Partially specific
- **Example:**
- "A person will play outside when the weather is sunny."
- Hypotheses are evaluated by checking whether they correctly classify the training examples.
- A good hypothesis should be consistent with the available data.

General-to-Specific Ordering

- Hypotheses can be arranged from **general to specific**.
- A **general hypothesis** accepts many possible examples.
- A **specific hypothesis** accepts only a limited number of examples.
- Training examples help refine the hypothesis.

Learning Process:

- Start with a possible hypothesis.
- Compare it with training examples.
- Generalise it when positive examples are missed.
- Make it more specific when negative examples are incorrectly accepted.

Importance and Applications

Importance
Helps machines learn concepts from examples.
Supports classification of new data.
Helps extract understandable knowledge and rules.
Forms the foundation for algorithms such as **Find-S** and **Candidate Elimination**.

Conclusion
Concept Learning enables a machine to learn a general rule from positive and negative examples. By representing and refining hypotheses, the system can classify new examples and extract useful knowledge from data.